AQA GCSE Physics
Topic – Forces
Section 1: Detailed Revision Notes

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1. Scalars and Vectors

Scalar Quantities

Scalars have **magnitude only** (size).

Examples:

- Distance
- Speed
- Mass
- Time
- Energy

Vector Quantities

Vectors have **magnitude and direction**.

Examples:

- Velocity
- Force
- Acceleration
- Displacement
- Weight

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2. Contact and Non-Contact Forces

Contact Forces

Forces that act when objects touch.

Examples:

- Friction
- Air resistance
- Tension
- Normal contact force

Non-Contact Forces

Forces that act at a distance.

Examples:

- Gravitational force
- Magnetic force
- Electrostatic force

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3. Resultant Force

The **resultant force** is the overall force acting on an object.

- If forces act in the same direction → add them
- If forces act in opposite directions → subtract them

If resultant force = 0 N → object is in equilibrium (balanced forces).

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4. Newton's First Law

If the resultant force on an object is zero:

- It remains at rest, or
- It continues moving at constant velocity.

This is called **inertia**.

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5. Newton's Second Law

Force is equal to mass multiplied by acceleration.

Correct formula

$F = m \times a$

Where:
F = force (newtons, N)
m = mass (kilograms, kg)
a = acceleration (metres per second squared, m/s^2)

Rearranged formulas:

$a = F \div m$

$m = F \div a$

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6. Newton's Third Law

For every action, there is an equal and opposite reaction.

If object A exerts a force on object B,
object B exerts an equal and opposite force on object A.

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7. Weight and Gravity

Weight is the force due to gravity.

Formula:

$W = m \times g$

Where:
W = weight (newtons, N)
m = mass (kg)
g = gravitational field strength (N/kg)

On Earth:

$g = 9.8$ N/kg

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8. Stopping Distance

Stopping distance = thinking distance + braking distance

Thinking distance increases with:

- Speed
- Tiredness
- Alcohol
- Distractions

Braking distance increases with:

- Speed
- Worn brakes
- Worn tyres
- Wet or icy roads

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9. Work Done and Energy Transfer

Work is done when a force causes movement.

Formula:

Work done = Force × Distance

$W = F \times d$

Where:
W = work done (joules, J)
F = force (N)
d = distance moved in direction of force (m)

1 joule = 1 newton metre

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10. Elastic Potential Energy

Energy stored when an object is stretched or compressed.

Formula:

Elastic potential energy = $0.5 \times k \times e^2$

$E = 0.5\ k\ e^2$

Where:
E = elastic potential energy (J)
k = spring constant (N/m)
e = extension (m)

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11. Hooke's Law

Extension is directly proportional to force (up to limit of proportionality).

Formula:

$F = k \times e$

Where:
F = force (N)
k = spring constant (N/m)
e = extension (m)

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12. Momentum

Momentum is mass × velocity.

Formula:

$p = m \times v$

Where:
p = momentum (kg m/s)

m = mass (kg)
v = velocity (m/s)

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13. Conservation of Momentum

In a closed system:

Total momentum before = Total momentum after

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

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14. Required Practical – Investigating Springs

Students must be able to:

- Measure extension
- Plot force vs extension graph
- Identify limit of proportionality
- Calculate spring constant

Gradient of graph = spring constant (k)

Common GCSE Exam Mistakes

- Using mass instead of weight in calculations
- Forgetting to convert grams to kilograms
- Not squaring extension in elastic potential energy formula
- Confusing speed and velocity
- Forgetting units

Key Formula List (Higher Tier)

$F = m \times a$
$W = m \times g$
Work done $= F \times d$
$F = k \times e$
$E = 0.5\ k\ e^2$
$p = m \times v$
Stopping distance = thinking + braking

End of Section 1 – Forces